

## Problem 4

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**Problem 1** Given  $y(t) = t - \frac{\pi}{3}$  and  $w(t) = \sin(t)$ . Find:

(a)  $y(w(t))$

**Explanation.**  $y(w(t)) = y(\sin(t)) = \sin(t) - \frac{\pi}{3}$

(b)  $w(y(t))$

**Explanation.**  $w(y(t)) = w\left(t - \frac{\pi}{3}\right) = \sin\left(t - \frac{\pi}{3}\right)$

(c)  $w\left(y\left(\frac{4\pi}{3}\right)\right)$

**Explanation.**  $w\left(y\left(\frac{4\pi}{3}\right)\right) = \sin\left(\frac{4\pi}{3} - \frac{\pi}{3}\right) = \sin(\pi) = 0$

(d)  $y(w(\frac{4\pi}{3}))$

**Explanation.**  $y\left(w\left(\frac{4\pi}{3}\right)\right) = \sin\left(\frac{4\pi}{3}\right) - \frac{\pi}{3} = \frac{-\sqrt{3}}{2} - \frac{\pi}{3}$

You should know values of  $\sin(x)$  and  $\cos(x)$  for all values found on the standard unit circle.